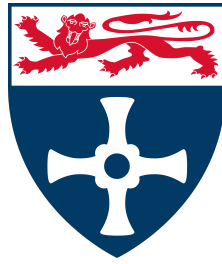


NEWCASTLE UNIVERSITY



SCHOOL OF COMPUTING

MSC DATA SCIENCE

CSC8639 — DISSERTATION

**An AI-Enabled Digital Twin for EV
Charging:
Behavioural Simulation, Policy
Evaluation,
and Pricing Optimisation**

Interim Report

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1 Introduction

The rapid global adoption of electric vehicles (EVs) represents one of the most consequential transitions in modern transportation infrastructure. The International Energy Agency reported that global EV stock surpassed 40 million vehicles in 2023, with projections indicating EVs could constitute over one-third of new vehicle sales by 2030 [1]. In the United Kingdom, the government’s Net Zero Strategy mandates the phase-out of new petrol and diesel vehicle sales by 2035, placing EV charging infrastructure at the centre of national decarbonisation policy [2].

The growth of EV adoption introduces significant operational challenges for charging networks. Demand is inherently heterogeneous, shaped by driver behaviour, time-of-day patterns, connector type preferences, and real-time grid carbon intensity. Static or rule-based approaches to charging management are poorly suited to this complexity, as they cannot adapt to dynamic demand, varying price signals, or grid constraints [7].

Digital twins, which are computational replicas of physical systems calibrated to reproduce real-world behaviour, have emerged as a promising paradigm for addressing these challenges. When implemented as agent-based simulations and calibrated on real historical data, digital twins allow policymakers to test interventions before deployment, without disrupting live infrastructure [4]. Agent-based modelling (ABM) is particularly well-suited to EV charging contexts, as heterogeneous driver agents can make autonomous decisions based on energy need, cost sensitivity, and station availability [3, 8].

Despite this potential, few studies have grounded digital twin simulations in real institutional charging data, and fewer still have applied reinforcement learning (RL) to autonomously discover optimal pricing policies [10, 12]. This project addresses both gaps by building a fully calibrated, AI-enabled digital twin of the EV charging network at Newcastle University’s Urban Sciences Building (USB), using 29,775 real historical charging sessions spanning 2021 to 2024.

To the authors’ knowledge, no prior study has grounded a digital twin simulation in real institutional EV charging data from a UK university campus, nor applied reinforcement learning to autonomously optimise pricing strategies in such a context.

2 Aim and Objectives

2.1 Aim

This project aims to develop, calibrate, and evaluate an AI-enabled agent-based digital twin of the EV charging network at Newcastle University’s Urban Sciences Building, using real historical session data to simulate heterogeneous charging behaviour, evaluate policy interventions under realistic uncertainty, and optimise pricing and incentive strategies for grid resilience and carbon reduction.

2.2 Objectives

- O1. To construct a reproducible data engineering pipeline that extracts, validates, and stores historical EV charging session data from the USB charging network, comprising 29,775 sessions across six chargers from March 2021 to July 2024, in a portable analytical database suitable for simulation calibration and downstream analysis.

- O2. To conduct a comprehensive exploratory data analysis of the charging session dataset, fitting statistical distributions to empirical charging behaviour including hourly arrival patterns, energy demand, session duration, and connector type preferences, and storing calibration parameters for use in the agent-based simulation.
- O3. To implement a calibrated agent-based simulation using the Mesa framework, in which heterogeneous EV driver agents reproduce key statistical properties of the historical dataset, including arrival distributions, energy demand, session duration, and charger utilisation, within a defined tolerance threshold.
- O4. To develop a scenario engine that stress-tests three policy interventions against the baseline simulation: EV adoption growth via scaled arrival rates; time-of-use (ToU) pricing via hour-varying price signals; and carbon-aware incentives via penalties on high-carbon charging periods, evaluating outcomes across four KPIs: grid load, station utilisation, average waiting time, and CO₂ emissions.
- O5. To produce a quantitative policy evaluation report comparing baseline and intervention scenarios across all KPIs, with statistical analysis across multiple simulation runs to account for stochastic variability.
- O6. (*Extension*) To implement a reinforcement learning policy optimisation module using Proximal Policy Optimisation (PPO) [9] via Stable-Baselines3 [15], wrapping the Mesa simulation as a Gymnasium-compatible environment [11], and demonstrating that the learned pricing policy outperforms handcrafted rules across the defined KPI set.

3 Overview of Progress

3.1 Literature Review

A targeted review of five papers has been conducted to ground the project methodology in current research. Papers span multi-agent EV charging simulation, digital twin frameworks for EV infrastructure, grid-integrated demand response, smart meter analytics for building energy benchmarking, and edge-AI reinforcement learning optimisation. Table 1 summarises the key contributions and their relevance.

Table 1: Summary of reviewed literature and relevance to this project.

Paper	Key Contribution	Relevance
Multi-Agent Game-Theoretic Model [3]	Utility-based EV agent with SOC urgency; dynamic pricing reduces peak congestion	Informs agent decision model and ToU pricing scenario
Digital Twin for EV Infrastructure [4]	ABM with PSO optimisation and GIS integration on a campus network	Validates Mesa as simulation framework; scenario engine design
Grid-Integrated DT Framework [5]	CPO incentive optimisation and demand response KPIs	Informs KPI selection: grid load, waiting time, CO ₂
Smart City DT for Building Energy [6]	Smart meter analytics and temporally-segmented benchmarking	Informs data pipeline design and metering data investigation
Edge-AI RL + AHP Framework [7]	RL with multi-criteria decision making; 23.5% wait time reduction reported	Motivates RL extension; AHP approach for KPI weighting

3.2 Data Engineering Pipeline

A complete, modular data engineering pipeline has been implemented in Python across four files: `ingestion/config.py`, `ingestion/storage.py`, `ingestion/scrapper.py`, and `ingestion/run_scraper.py`. The pipeline extracts 30-minute interval energy data from Newcastle University’s metering portal using authenticated POST requests, with parallel execution via `ThreadPoolExecutor` and a DuckDB-backed checkpoint table enabling fault-tolerant, resumable scraping.

All project data is stored in a single portable DuckDB analytical database (`ev_twin.duckdb`). Table 2 describes the database schema.

Table 2: DuckDB database tables and their purpose.

Table	Rows	Source	Purpose
<code>charging_sessions</code>	29,775	Supervisor CSV	Primary simulation calibration data
<code>metering_raw</code>	28,728,144	NCL metering portal	Building energy data (investigated)
<code>simulation_params</code>	7 sets	Derived from EDA	Agent calibration parameters
<code>scrape_checkpoint</code>	598,596	System-generated	Fault-tolerant resume mechanism

Source code is version-controlled at: <https://github.com/NguyenIslandBoy/ev-digital-twin>

3.3 Metering Data Investigation

To assess whether building switchboard meter data (`USB_E_SwitchMA` and `USB_E_SwitchMB`) could be integrated with charging session data for grid load validation, a systematic data quality investigation was conducted. Charging sessions were aggregated to 30-minute windows and joined on timestamp across 57,696 matched intervals.

Two critical findings emerged. First, the Pearson correlation between aggregated session energy and switchboard meter readings was $r = -0.086$, indicating no meaningful relationship. Second, monthly analysis of distinct meter values per interval revealed that 34 of 40 months contained a single repeated value across all 48 daily slots, which is indicative of metering system collection failures rather than genuine interval consumption data. Only 6 months yielded real interval-level readings, representing less than 15% of the study period. This was confirmed by the project proposer, who verified these are the only meters capturing EV-related load in the USB building.

Decision: `usb_merged_final_data.csv` is used as the sole data source. Grid load KPIs will be derived from simulation output rather than real meter readings. The metering investigation is documented as a data quality finding demonstrating analytical rigour.

3.4 Exploratory Data Analysis

A comprehensive EDA was conducted on 29,775 charging sessions in `notebooks/01_eda.ipynb`, following established approaches for characterising empirical EV charging behaviour [13]. Key findings that directly inform agent behaviour parameters are summarised in Table 3.

Table 3: Key EDA findings and their simulation implications.

Finding	Detail	Simulation Implication
Bimodal arrival pattern	Peaks at 11:00–12:00 and 14:00–16:00, consistent with lecture schedule	Empirical hourly arrival rates; Poisson sampling per hour
Weekday vs weekend split	68.93% weekday sessions; weekend peak shifts to 10:00–11:00	Separate arrival distributions by day type
Connector type distribution	IEC_62196_T2_COMBO: 82.07%, CHADEMO: 14.32%, IEC_62196_T2: 3.61%	Categorical sampling per agent
Energy distribution	Mean 30.4 kWh; flat plateau 5–20 kWh; Gamma fit poor (AIC-selected but mixture distribution evident)	Empirical sampling per connector type
Duration distribution	Gamma fit confirmed (AIC-selected); median 0.64 hrs; capped at 4 hrs (99.9th percentile)	Gamma sampling per connector type
Charger utilisation	6 chargers; 5000198 and 5000199 handle $\sim 48\%$ of sessions	Per-charger capacity modelled explicitly
Seasonal pattern	Spring highest, Summer lowest (term-time effect)	Season-aware arrival scaling in scenario engine

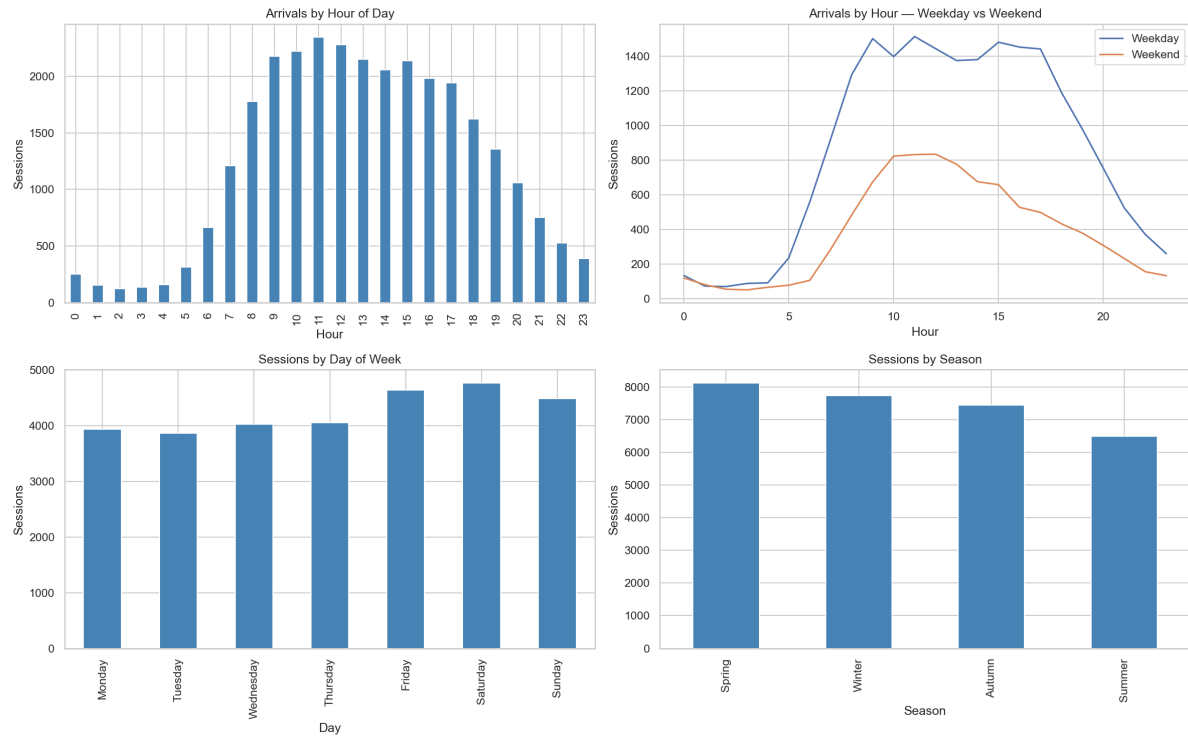


Figure 1: Hourly arrival distributions across all sessions (left), and weekday vs weekend comparison (right). Bimodal peaks at 11:00–12:00 and 14:00–16:00 are consistent with the USB building lecture schedule.

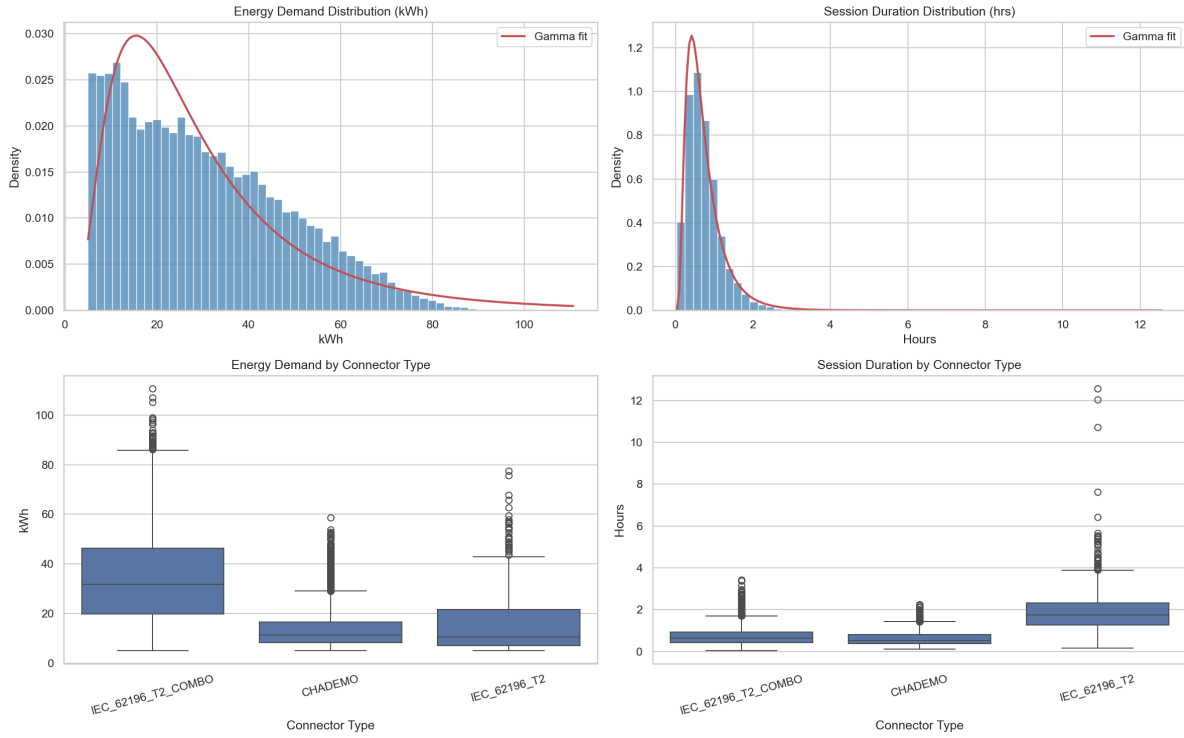


Figure 2: Energy demand and session duration distributions with Gamma fits (AIC-selected), and per-connector type boxplots. The Gamma fit is poor for energy due to a flat plateau between 5–20 kWh, motivating empirical sampling; the Gamma fit is good for duration.

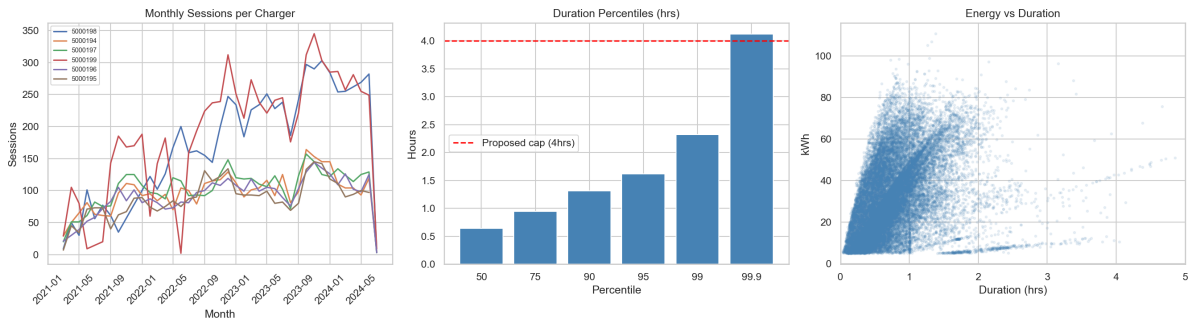


Figure 3: Monthly session counts per charger (left), duration percentile analysis with proposed 4-hour cap (centre), and energy vs duration scatter plot (right).

3.5 Key Design Decisions

Table 4 summarises the major technical decisions made and their justifications.

Table 4: Key technical design decisions and justifications.

Decision	Choice	Justification
Storage	DuckDB	Zero server setup, columnar analytics, portable, native pandas integration
Simulation framework	Mesa [14, 8]	Python-native ABM, appropriate for 6-charger network scale, reproducible
Energy demand modelling	Empirical sampling per connector type	Lognormal fit visually poor; flat plateau 5–20 kWh indicates mixture distribution
Duration modelling	Gamma per connector type (AIC-selected)	Gamma fit visually confirmed; separate params per connector captures behavioural heterogeneity
Metering integration	Not integrated	Collection failures in 34/40 months; $r = -0.086$; confirmed by supervisor
RL framework	Stable-Baselines3 PPO on Colab [15, 10]	Production-grade library; PPO stable for continuous action spaces; Colab A100 for compute

4 Project Plan

4.1 Phases and Timeline

Table 5 summarises the five project phases, linked to the objectives defined in Section 2.2.

Table 5: Project phases, tasks, and target completion dates.

Phase	Key Tasks	Objectives Target
Phase 1: Data Engineering	Pipeline, DuckDB storage, metering investigation, EDA	O1, O2 Complete (May 2026)
Phase 2: Mesa Simulation	Agent design, charger resource model, calibration validation	O3 June 2026
Phase 3: Scenario Engine	ToU pricing, EV adoption growth, carbon incentive scenarios, KPI evaluation	O4, O5 Mid July 2026
Phase 4: RL Extension	Gym environment wrapper, PPO training on Colab, policy comparison	O6 Late July 2026
Phase 5: Writing and Submission	Dissertation, source code cleanup, documentation, poster, oral presentation	All 10–13 Aug 2026

4.2 Milestones

Table 6: Project milestones aligned with official submission deadlines.

Date	Milestone
5 June 2026	Interim report submitted
30 June 2026	Calibrated Mesa simulation validated against historical data
20 July 2026	All three what-if scenarios evaluated with KPI report
29 July 2026	Research poster submitted (10 pts)
10 August 2026	Dissertation and source code submitted
13 August 2026	Oral presentation evidence submitted

4.3 Risks

Table 7: Project risk register with likelihood, impact, and mitigation strategies.

Risk	Likelihood	Impact	Mitigation
Mesa simulation fails to calibrate within tolerance	Medium	High	Use KS-test for distribution comparison; fall back to empirical session replay if needed
RL agent fails to converge	Medium	Medium	RL is an extension; core deliverable is the scenario engine; reduce state space if needed
Dissertation writing underestimated	High	High	Budget 5 weeks minimum; begin writing in parallel with simulation development in June
Session cookie expiry during scrape	Low	Low	Checkpoint table enables resume; scrape is already complete

4.4 Gantt Chart

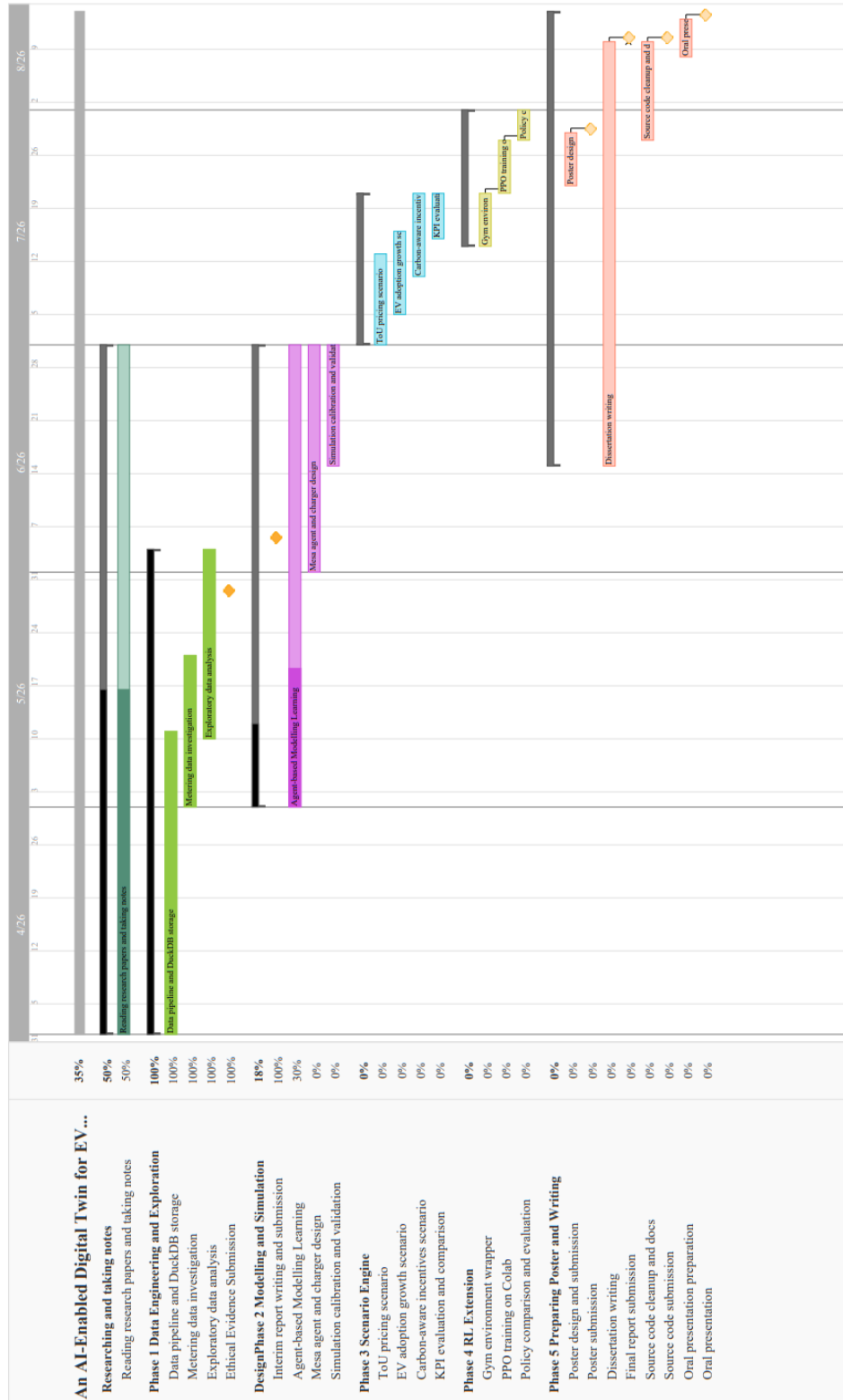


Figure 4: Gantt chart of the proposed project. Tasks are colour-coded by phase. Progress percentages are shown at the end of each bar. Milestones are shown as diamonds.

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0. Proposal name																											
An AI-Enabled Digital Twin for EV Charging: Behavioural Simulation, Policy Evaluation, and Pricing Optimisation - CSC8639 MSc Dissertation, Hoang Nguyen Lai (250539846)																											
1. Description of the data																											
1.1 Type of study <i>This project develops a calibrated agent-based digital twin of the EV charging network at Newcastle University's Urban Sciences Building (USB), using historical charging session data to simulate driver behaviour, evaluate policy interventions, and optimise pricing strategies via reinforcement learning.</i> 1.2 Types of data <i>The project uses two categories of primary data. First, quantitative charging session records comprising 29,775 sessions from 2021 to 2024, provided by the project proposer, covering session timestamps, energy consumed in kWh, session duration, connector type, charger ID, and carbon intensity. Second, quantitative building metering data comprising 28.7 million 30-minute interval energy readings collected via an automated scraper from Newcastle University's metering portal. The project additionally generates derived quantitative data including simulation calibration parameters, trained reinforcement learning model weights, and analysis outputs.</i> 1.3 Format and scale of the data <table> <tr> <th>Asset</th><th>Format</th><th>Scale</th><th>Long-Term Validity</th></tr> <tr> <td>Charging sessions</td><td>CSV</td><td>~4 MB</td><td>Open standard; readable by any tool indefinitely</td></tr> <tr> <td>Building metering data</td><td>DuckDB (.duckdb)</td><td>~1.39 GB</td><td>Plain text; human-readable and fully reproducible</td></tr> <tr> <td>Simulation parameters</td><td>JSON</td><td><1 MB</td><td>Plain text; human-readable and fully reproducible</td></tr> <tr> <td>Source code and notebooks</td><td>.py, .ipynb</td><td><10 MB</td><td>Python 3.14; dependencies pinned in requirements.txt</td></tr> <tr> <td>RL model weights</td><td>.zip (Stable-Baseline3)</td><td><50 MB</td><td>Standard ML serialisation format</td></tr> </table> <i>All formats are open, non-proprietary, and suitable for long-term reuse.</i>				Asset	Format	Scale	Long-Term Validity	Charging sessions	CSV	~4 MB	Open standard; readable by any tool indefinitely	Building metering data	DuckDB (.duckdb)	~1.39 GB	Plain text; human-readable and fully reproducible	Simulation parameters	JSON	<1 MB	Plain text; human-readable and fully reproducible	Source code and notebooks	.py, .ipynb	<10 MB	Python 3.14; dependencies pinned in requirements.txt	RL model weights	.zip (Stable-Baseline3)	<50 MB	Standard ML serialisation format
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2. Data collection / generation																											

2.1 Methodologies for data collection / generation <i>Charging session data was provided by Ms Sanchari Deb under institutional supervisory data sharing; the raw file is preserved unchanged. Metering data was collected via an automated Python scraper with a DuckDB checkpoint table auditing all 598,596 requests. No clinical or survey collection standards apply.</i> 2.2 Data quality and standards <i>Raw data is never edited directly; all transformations are applied via versioned SQL scripts. A data quality investigation of the metering data identified systemic collection failures in 34 of 40 months (Pearson $r = -0.086$); this is documented in the EDA notebook with supporting queries. All code is version-controlled on GitHub.</i>
3. Data management, documentation and curation
3.1 Managing, storing and curating data. <i>Data is stored locally with weekly external drive backups. Code and notebooks are version-controlled at https://github.com/NguyenIslandBoy/ev-digital-twin. Raw data is excluded from the repository via .gitignore.</i> 3.2 Metadata standards and data documentation <i>Documentation includes: README.md (project structure and execution), requirements.txt and pyproject.toml (versioned dependencies), inline docstrings, and sequentially executable notebooks with narrative markdown. Calibration parameters are stored as structured JSON in DuckDB.</i> 3.3 Data preservation strategy and standards <i>Code and documentation retained on GitHub indefinitely. Raw data and database retained locally for minimum 12 months post-submission per Newcastle University's Research Data Management Policy. Raw CSV not shared publicly without supervisor approval.</i>
4. Data security and confidentiality of potentially disclosive information
4.1 Formal information/data security standards <i>Compliant with Newcastle University's Information Security Policy and Research Data Management Policy. No personal data is processed; full ISO 27001 compliance is not required.</i> 4.2 Main risks to data security <i>No personal data is held. Auth IDs are anonymised integers. Key risks: (1) accidental credential exposure for mitigated by excluding config files from version control; (2) repository access during development for mitigated by keeping repository private until submission with two-factor authentication; (3) data loss for mitigated by weekly external backups.</i>
5. Data sharing and access
5.1 Suitability for sharing <i>Code, notebooks, simulation parameters, and model weights will be made publicly available on GitHub after submission. Raw charging session CSV and metering data are not suitable for open sharing without supervisor approval.</i> 5.2 Discovery by potential users of the research data <i>The public GitHub repository will include a detailed README and be discoverable via GitHub search. The repository URL will be cited in the dissertation.</i> 5.3 Governance of access <i>Decisions on sharing the raw dataset rest with Dr Jichun Li. All other outputs will be freely available under the repository licence.</i>

5.4 The study team's exclusive use of the data <i>The GitHub repository will remain private until dissertation assessment is confirmed complete, expected by September 2026. After that point, the repository will be made public. The raw charging session dataset will remain under the governance of the project proposer indefinitely, with no defined release date for open access.</i>	
5.5 Restrictions or delays to sharing, with planned actions to limit such restrictions <i>No consent-related restrictions apply. The only restriction is the institutional nature of the raw CSV, managed by excluding it from the public repository while sharing all derived outputs freely.</i>	
5.6 Regulation of responsibilities of users <i>External users requesting the raw dataset must obtain approval from Dr Jichun Li and agree to use data for academic purposes only, not attempt re-identification, and cite the source appropriately. All other outputs are governed by the repository's open-source licence.</i>	
6. Responsibilities <i>Apart from the PI, who is responsible at your organisation/within your consortia for:</i> <i>• Data management, quality assurance, security, documentation: Hoang Nguyen Lai (Postgraduate student)</i> <i>• Technical Supervision: Dr Jichun Li</i> <i>• Dataset governance: Ms Sanchari Deb (project proposer)</i>	
7. Relevant institutional, departmental or study policies on data sharing and data security	
Policy	URL or Reference
Data Management Policy & Procedures	https://www.ncl.ac.uk/mediav8/our-research/research-governance-policies/ResearchDataManagementPolicyandCoGP.pdf
Data Security Policy	https://www.ncl.ac.uk/data-protection/data-protection-policy/data-protection-policy-june-2020/
Other	
8. Author of this Data Management Plan (Name) and, if different to that of the Principal Investigator, their telephone & email contact details	
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